

SAT Math Mastery · Week TP1K**SAT Trick Pack 1 — Part K****Time** 120 minutes: 90 min solving + 30 min self-check and error log**Topics** coefficient sums · mixtures · piecewise break-even · asymptote vs. maximum · intersecting lines · angle-sum identities · linear evaluation · repeated roots · absolute-value vertex · multiplicity · quadratic symmetry · compound percent · average speed · factor arrangements · transformations**Name / Date** _____**1 · SKILLS & IDEAS**

- S1.** Find the sum of the coefficients of an equivalent form by evaluating the original expression at $x = 1$.
- S2.** Set up a mixture system in one consistent unit and answer for the variable actually asked.
- S3.** Compare piecewise plans at the breakpoints to find the right interval, then solve one linear equation.
- S4.** Tell an attained maximum from an asymptotic limit on a restricted domain.
- S5.** Use supplementary angles at intersecting lines to fix every angle, then test which sums are possible.
- S6.** Recognise $\sin A \cos B + \cos A \sin B$ and $\cos A \cos B - \sin A \sin B$ as angle-sum formulas.
- S7.** Combine fractions before multiplying by a slope; evaluate a line at a combined input.
- S8.** Group a parameter cubic into a product; a repeated root leaves two distinct solutions.
- S9.** A line meets a V-shaped graph once only through the vertex.
- S10.** Use multiplicity (touch vs. cross), sign intervals and end behaviour to judge statements.
- S11.** Equal outputs fix the axis; compare vertex height with the y -intercept to get direction and signs.
- S12.** Chain percent multipliers whose second rate depends on the first; reject roots that contradict the story.
- S13.** Total distance over total time (rest included), then convert units with one fraction.
- S14.** Move shared constants between the outer factor and the binomials to compare arrangements.
- S15.** Undo a transformation of a function from one point to find a missing coefficient.

2 · REQUIRED FACTS

- The sum of the coefficients of a polynomial $p(x)$ is $p(1)$.
- A quantity that only approaches a value (a horizontal asymptote) never attains it; a maximum must actually be reached.
- Angles on a straight line sum to 180° ; vertical angles are equal.
- $\sin(A + B) = \sin A \cos B + \cos A \sin B$ and $\cos(A + B) = \cos A \cos B - \sin A \sin B$; $\sin 150^\circ = \frac{1}{2}$, $\cos 150^\circ = -\frac{\sqrt{3}}{2}$.
- A factor $(x - r)^m$ gives a zero at r : the graph crosses if m is odd and touches if m is even.
- $y = |x - h| + v$ has vertex (h, v) ; a line shallower than the arms meets it twice, once or never according to its position relative to the vertex.
- If $f(p) = f(q)$ for a quadratic, the axis is $x = \frac{p+q}{2}$; a vertex lower than the y -intercept means the parabola opens upward.
- Raising by $r\%$ multiplies by $1 + \frac{r}{100}$; lowering by $s\%$ multiplies by $1 - \frac{s}{100}$.
- $1 \text{ km/h} = \frac{1000}{3600} \text{ m/s} = \frac{5}{18} \text{ m/s}$; average speed $= \frac{\text{total distance}}{\text{total time}}$ including rests.
- If $g(x) = a f(x + h) + k$ then $g(t)$ is computed from $f(t + h)$: shift first, then scale, then move.

3 · REQUIRED FORMULAS

$$p(1) = \text{sum of coefficients}, \quad x_{\text{axis}} = \frac{p+q}{2}, \quad \bar{v} = \frac{d_{\text{total}}}{t_{\text{total}}}, \quad m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\sin(A+B) = \sin A \cos B + \cos A \sin B, \quad \cos(A+B) = \cos A \cos B - \sin A \sin B, \quad \left(1 + \frac{r}{100}\right)\left(1 - \frac{s}{100}\right) = \text{net factor}$$

4 · MINI THEORY SHEET

Evaluate, do not expand. The sum of all coefficients of an expression is its value at $x = 1$; divide or add exactly as the original does. The constant term is its value at $x = 0$.

Mixtures and units. Convert every price or rate to the same unit first, write one equation for the totals and one for the cost (or concentration), then answer for the quantity the question names.

Piecewise plans. Evaluate both plans at each breakpoint, see where one overtakes the other, and solve a single linear equation inside that interval only; check the answer lies inside it.

Maximum vs. limit. For a decreasing exponential on $t \geq 0$ the maximum is the starting value; for an increasing one that levels off, the asymptote is never reached, so there is no maximum.

Parameters and repeated roots. Group parameter terms together, factor out the common factor, and read the roots. A repeated root counts as one distinct solution.

Percent chains. Write every change as a multiplier, multiply, set equal to the net factor, solve the quadratic and discard roots that break a stated condition.

Speed shortcuts.

Situation	Shortcut
sum of coefficients of a product/quotient	substitute $x = 1$ into the original form
mixture with mixed units	convert to one unit, then two equations; re-read the question for the asked variable
two piecewise plans	test both at the breakpoint to pick the interval; one linear equation
$c \pm ab^t$ on $t \geq 0$	decreasing: maximum at $t = 0$; increasing to c : no maximum
two intersecting lines, algebraic angles	acute + obtuse = 180° ; list every possible pair sum
product of sines and cosines of two angles	name the angles, find $A + B$, apply the sum formula
line through two fractional points	single fraction for Δx and Δy ; then point-slope
cubic with parameter k	group, factor, set k equal to an existing root
$ ax - k + k$ against a line	vertex in terms of k must lie on the line
even vs. odd multiplicity	touch vs. cross; test one point to refute “always” claims
equal outputs of a quadratic	axis at the midpoint; vertex vs. intercept gives the sign of a
up $r\%$ then down $(r - c)\%$	$(1 + \rho)(1 + c/100 - \rho) = \text{net}$; keep the root that fits the story
trip with a rest stop	total distance $\div$ total time, rest included; convert with $\frac{5}{18}$
$k(\dots)(\dots)$ arrangements	push shared constants between factors and compare the sums
$g(x) = af(x + h) + k$ with one point	substitute the point, undo k , divide by a , read $f(x + h)$

5 · GUIDED EXAMPLES

Example 1 — Sum of coefficients: substitute $x = 1$

The expression $\frac{(x-2)^3(2x+1)+5}{3}$ is equivalent to $ax^4 + bx^3 + cx^2 + dx + e$. Find $a + b + c + d + e$.

The sum of the coefficients is the value at $x = 1$: $\frac{(-1)^3(3)+5}{3} = \frac{2}{3}$. No expansion needed.

Example 2 — A line meets a V once only at the vertex

$|x - 4| + 1 = \frac{x}{2} + c$ has exactly one solution. Find c .

The V has vertex $(4, 1)$ and arms of slope ± 1 , steeper than the line. One intersection only if the line passes through the vertex: $1 = \frac{4}{2} + c \Rightarrow c = -1$.

Example 3 — Spot the angle-sum formula

Evaluate $\sin 20^\circ \cos 70^\circ + \cos 20^\circ \sin 70^\circ$.

This is $\sin(20^\circ + 70^\circ) = \sin 90^\circ = 1$. Never expand the four values.

6 · PRACTICE SET — 15 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. SAT CHALLENGE · NO CALCULATOR NEEDED
The expression $\frac{(2x-3)^4(x+2)-7}{5}$ is equivalent to $ax^5 + bx^4 + cx^3 + dx^2 + ex + f$, where a, b, c, d, e and f are constants. What is the value of $a + b + c + d + e + f$?

(A) 31

(B) $-\frac{4}{5}$ ✓(C) $\frac{3}{5}$

(D) -4

- The sum of all the coefficients of a polynomial $p(x)$ equals $p(1)$, because every power of 1 is 1.
- The expression is already equal to the polynomial, so evaluate it at $x = 1$: $(2 \cdot 1 - 3)^4 = (-1)^4 = 1$ and $(1 + 2) = 3$.
- Numerator: $1 \cdot 3 - 7 = -4$. Dividing by 5: $-\frac{4}{5}$.
- $a + b + c + d + e + f = -\frac{4}{5}$. (The other options are: $\frac{3}{5}$ forgets the -7 , -4 forgets the division by 5, and 31 is the constant term f , the value at $x = 0$.)

Answer: (B)

2. SAT CHALLENGE · DESMOS
A coffee shop blends Ethiopian beans, which cost \$18 per kilogram, with Colombian beans, which cost \$11 per 500 grams. The blend has a total mass of 36 kilograms, and its cost works out to \$20.50 per kilogram. How many kilograms of Ethiopian beans are in the blend?

(A) 18

(B) 27

(C) 13.5 ✓

(D) 22.5

- Put both prices in dollars per kilogram: Colombian beans cost \$11 per 0.5 kg, which is \$22 per kilogram.
- Let x and y be the kilograms of Ethiopian and Colombian beans. Mass: $x + y = 36$. Cost: $18x + 22y = 20.50(36) = 738$.
- Substitute $x = 36 - y$: $18(36 - y) + 22y = 738 \Rightarrow 648 + 4y = 738 \Rightarrow y = 22.5$, so $x = 13.5$.
- Check: $18(13.5) + 22(22.5) = 243 + 495 = 738$.
- The question asks for the Ethiopian beans: 13.5 kg. (22.5 is the Colombian mass.)

Answer: (C)

3.

SAT CHALLENGE · CALCULATOR

Plan P costs \$25 per month for up to 4 gigabytes (GB) of data, plus \$6 for each additional GB. Plan Q costs \$14 per month for up to 2 GB of data, plus \$0.008 for each additional megabyte (MB). Charges for additional data are prorated, and 1 GB = 1000 MB. For what amount of data, in MB, do the two plans cost the same?

1. Use one unit. Plan Q's extra charge is \$0.008 per MB, which is \$8 per GB.
2. Compare at the breakpoints. At 2 GB: P costs \$25, Q costs \$14. At 4 GB: P costs \$25, Q costs $14 + 8(2) = \$30$. Q starts cheaper and is more expensive by 4 GB, so the plans meet between 2 and 4 GB, where P is still the flat \$25.
3. Solve in that interval: $14 + 8(g - 2) = 25 \Rightarrow 8(g - 2) = 11 \Rightarrow g = 3.375$ GB.
4. Convert: $3.375 \times 1000 = 3375$ MB.
5. (Solving with P's per-GB charge, $14 + 8(g - 2) = 25 + 6(g - 4)$, gives $g = 1.5 < 2$, outside the interval, so it is not valid.)

Answer: 3375

4.

SAT CHALLENGE · NO CALCULATOR NEEDED

The temperature T , in degrees Fahrenheit, of a sample t minutes after it is placed in a chamber is modeled by one of the functions below for $t \geq 0$. For which model is 85 the greatest temperature the sample actually reaches?

(A) $T = 25 + 60(0.8)^t$ ✓

(B) $T = 85 - 60(0.8)^t$

(C) $T = 40 + 45(1.05)^t$

(D) $T = 25 + 85(0.8)^t$

1. Test each model on $t \geq 0$ and ask whether 85 is reached, not merely approached.
2. The model $T = 85 - 60(0.8)^t$ increases toward 85 but is always below it: 85 is a horizontal asymptote, never attained. No maximum.
3. The model $T = 25 + 60(0.8)^t$ decreases, so its greatest value is at $t = 0$: $25 + 60 = 85$. Attained. ✓
4. The model $T = 40 + 45(1.05)^t$ grows without bound, and $T = 25 + 85(0.8)^t$ has maximum $25 + 85 = 110$ at $t = 0$.
5. Only $T = 25 + 60(0.8)^t$ has 85 as an attained greatest value.

Answer: (A)

5.

SAT CHALLENGE · NO CALCULATOR NEEDED

Two lines intersect and form four angles. The acute angles each measure $(2k + 4)^\circ$, where k is a constant. Each obtuse angle measures 4 degrees more than 3 times the measure of an acute angle. Which of the following CANNOT be the sum, in degrees, of the measures of two of the four angles?

(A) 272

(B) 88

(C) 180

(D) 224 ✓

1. An acute angle and an obtuse angle at an intersection are supplementary: $(2k + 4) + [3(2k + 4) + 4] = 180$.
2. $4(2k + 4) + 4 = 180 \Rightarrow 8k + 20 = 180 \Rightarrow k = 20$.
3. Acute angles: $2(20) + 4 = 44^\circ$. Obtuse angles: $3(44) + 4 = 136^\circ$. Check: $44 + 136 = 180$.
4. Sums of two of the four angles: acute + acute = 88; acute + obtuse = 180; obtuse + obtuse = 272.
5. 224 is not among them, so it cannot be a sum.

Answer: (D)

6.

SAT CHALLENGE · NO CALCULATOR NEEDED

In triangle ABC , the measure of angle A is $(3x + 11)^\circ$, the measure of angle B is $(2x + 4)^\circ$, and the measure of angle C is 30° . What is the value of

$$(\sin A \cos B + \cos A \sin B)(\cos A \cos B - \sin A \sin B)?$$

(A) $-\frac{1}{4}$ (B) $\frac{1}{4}$ (C) $-\frac{\sqrt{3}}{4}$ ✓(D) $\frac{\sqrt{3}}{4}$

1. The angles of a triangle sum to 180° , so $A + B = 180^\circ - 30^\circ = 150^\circ$. (Solving, $5x + 15 = 150$, $x = 27$, but the value of x is not needed.)
2. The first factor is $\sin(A + B) = \sin 150^\circ = \frac{1}{2}$.
3. The second factor is $\cos(A + B) = \cos 150^\circ = -\frac{\sqrt{3}}{2}$.
4. Product: $\frac{1}{2} \cdot \left(-\frac{\sqrt{3}}{2}\right) = -\frac{\sqrt{3}}{4}$. (Using $\sin 30^\circ \cos 30^\circ$ instead would give the wrong sign.)

Answer: (C)

7.

SAT CHALLENGE · NO CALCULATOR NEEDED

The linear function f satisfies $f\left(\frac{2}{3}\right) = \frac{1}{4}$ and $f\left(\frac{8}{3}\right) = \frac{13}{4}$. What is the value of $f\left(\frac{1}{2} + \frac{3}{5}\right)$?

1. Slope: $m = \frac{\frac{13}{4} - \frac{1}{4}}{\frac{8}{3} - \frac{2}{3}} = \frac{3}{2}$.

2. Combine the input first: $\frac{1}{2} + \frac{3}{5} = \frac{5}{10} + \frac{6}{10} = \frac{11}{10}$.

3. Point-slope from $\left(\frac{2}{3}, \frac{1}{4}\right)$: $f(x) = \frac{1}{4} + \frac{3}{2}\left(x - \frac{2}{3}\right)$.

4. $x - \frac{2}{3} = \frac{11}{10} - \frac{2}{3} = \frac{33 - 20}{30} = \frac{13}{30}$, so $f\left(\frac{11}{10}\right) = \frac{1}{4} + \frac{3}{2} \cdot \frac{13}{30} = \frac{1}{4} + \frac{13}{20} = \frac{18}{20} = \frac{9}{10}$.

5. Answer: $\frac{9}{10}$ (or 0.9). Note f of a sum is not the sum of the f -values.

Answer: $\frac{9}{10}$

8.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$x^3 + (3 - k)x^2 - (10 + 3k)x + 10k = 0$$

In the equation above, k is a constant. The equation has exactly two distinct real solutions for exactly two values of k . What is the sum of these two values of k ?

1. Group the terms with k and the terms without: $(x^3 + 3x^2 - 10x) - k(x^2 + 3x - 10) = 0$.

2. Factor $x^3 + 3x^2 - 10x = x(x^2 + 3x - 10)$, so the equation is $(x^2 + 3x - 10)(x - k) = 0$.

3. $x^2 + 3x - 10 = (x + 5)(x - 2)$, so the equation is $(x + 5)(x - 2)(x - k) = 0$ with solutions -5 , 2 and k .

4. There are two distinct solutions only when k repeats one of the fixed roots: $k = 2$ or $k = -5$. (Otherwise there are three.)

5. Sum: $2 + (-5) = -3$.

Answer: -3

9.

SAT CHALLENGE · DESMOS

$$|2x - k| + k = x + 6$$

In the equation above, k is a constant. The equation has exactly one solution. What is the value of k ?

1. Graph $y = |2x - k| + k$: a V with vertex $\left(\frac{k}{2}, k\right)$ and arms of slope ± 2 . The line $y = x + 6$ has slope 1, shallower than both arms.
2. A shallower line meets such a V twice if it passes above the vertex, never if it passes below, and exactly once only if it passes through the vertex.
3. Put the vertex on the line: $k = \frac{k}{2} + 6 \Rightarrow \frac{k}{2} = 6 \Rightarrow k = 12$.
4. Check: $|2x - 12| + 12 = x + 6$ gives $x = 6$ from the right branch, and the left branch gives $x = 6$ as well (the vertex), so there is one solution.

Answer: 12

10.

SAT CHALLENGE · NO CALCULATOR NEEDED

The polynomial function p is defined by $p(x) = -(x + 3)^2(x - 1)^3(x - 4)^2$. Consider the following statements.

- I. $p(x) \geq 0$ for all $x < 1$.
- II. The graph of $y = p(x)$ crosses the x -axis at $x = 4$.
- III. $p(x) \rightarrow \infty$ as $x \rightarrow -\infty$.

Which of the statements must be true?

(A) I only

(B) I and II only

(C) I and III only ✓

(D) II and III only

1. Multiplicities: $(x + 3)^2$ and $(x - 4)^2$ are even, so the graph touches the axis at -3 and 4 without crossing. $(x - 1)^3$ is odd, so the graph crosses at 1 .
2. I: for $x < 1$, $(x - 1)^3 < 0$, so $-(x - 1)^3 > 0$, while the squares are ≥ 0 . Hence $p(x) \geq 0$, with $p(-3) = 0$. True. (The sign does not change at $x = -3$, but $p(-3) = 0$, so $\geq$ is needed.)
3. II: $x = 4$ has even multiplicity, so the graph touches and does not cross. False.
4. III: the degree is $2 + 3 + 2 = 7$ (odd) and the leading coefficient is negative, so $p(x) \rightarrow +\infty$ as $x \rightarrow -\infty$ (and $-\infty$ as $x \rightarrow \infty$). True.
5. I and III only.

Answer: (C)

11.

SAT CHALLENGE · NO CALCULATOR NEEDED

The quadratic function $h(x) = ax^2 + bx + c$, where a , b and c are constants, satisfies $h(-3) = h(1)$, $h(0) > h(-1)$ and $h(-1) > 0$. Consider the following statements.

- I. $b > 0$
- II. $c > 0$
- III. The equation $h(x) = 0$ has two distinct real solutions.

Which of the statements must be true?

(A) I only

(B) I and II only ✓

(C) II and III only

(D) I, II, and III

1. $h(-3) = h(1)$ puts the axis at $x = \frac{-3+1}{2} = -1$, so the vertex is $(-1, h(-1))$ and $-\frac{b}{2a} = -1$, i.e. $b = 2a$.
2. $h(0) = c$ is the y -intercept and $h(-1)$ is the vertex height. Since $h(0) > h(-1)$ the vertex is the lowest point, so the parabola opens upward: $a > 0$.
3. I: $b = 2a > 0$. True.
4. II: $c = h(0) > h(-1) > 0$. True.
5. III: the minimum value $h(-1)$ is positive and the parabola opens upward, so the graph never reaches the x -axis: no real solutions. False.
6. I and II only.

Answer: (B)

12.

SAT CHALLENGE · CALCULATOR

A store raises the price of a jacket by r percent. Later, it reduces the new price by $(r - 13)$ percent. After both changes, the final price is 10% higher than the original price. What is the value of r ?

1. Write each change as a multiplier. With $\rho = \frac{r}{100}$: increase by $r\%$ gives $1 + \rho$; reduce by $(r - 13)\%$ gives $1 - (\rho - 0.13) = 1.13 - \rho$. (The percent is reduced by 13 before converting, not multiplied.)
2. Net factor 1.10: $(1 + \rho)(1.13 - \rho) = 1.10$.
3. Expand: $1.13 - \rho + 1.13\rho - \rho^2 = 1.10 \Rightarrow \rho^2 - 0.13\rho - 0.03 = 0$.
4. Factor: $(\rho - 0.25)(\rho + 0.12) = 0$, so $\rho = 0.25$ or $\rho = -0.12$.
5. A raise means $\rho > 0$ (and $r > 13$ for a reduction). Reject $\rho = -0.12$. Then $r = 25$.
6. Check: $1.25 \times (1 - 0.12) = 1.25 \times 0.88 = 1.10$.

Answer: 25

13.

SAT CHALLENGE · CALCULATOR

A cyclist rides a 90-kilometer route. She covers the first 60% of the distance at an average speed of 27 kilometers per hour, rests for 30 minutes, and then covers the rest of the route at an average speed of 24 kilometers per hour. What was her average speed for the whole trip, including the rest, in meters per second?

(A) 22.5

(B) 6.25 ✓

(C) 7.08

(D) 7.14

1. Distances: first part $0.6 \times 90 = 54$ km, second part 36 km.

2. Times: $\frac{54}{27} = 2$ h riding, 0.5 h resting, $\frac{36}{24} = 1.5$ h riding. Total time = $2 + 0.5 + 1.5 = 4$ h.

3. Average speed = $\frac{90}{4} = 22.5$ km/h.

4. Convert: $22.5 \times \frac{1000}{3600} = 22.5 \times \frac{5}{18} = 6.25$ m/s.

5. (Distractors: 7.14 leaves out the rest, 7.08 averages the two speeds, 22.5 is not converted.)

Answer: (B)

14.

SAT CHALLENGE · NO CALCULATOR NEEDED

The polynomial $12x^4 - 84x^2 + 144$ can be written in the form $(ax^2 + b)(cx^2 + d)$, where a , b , c and d are integers with $a > 0$ and $c > 0$. What is the least possible value of $a + b + c + d$?

(A) -27

(B) -22

(C) -18

(D) -38 ✓

1. Factor out the common 12: $12(x^4 - 7x^2 + 12) = 12(x^2 - 3)(x^2 - 4)$. Neither $x^2 - 3$ nor $x^2 - 4$ can be rewritten with a different pair of roots, so every valid form is $\alpha(x^2 - 3) \cdot \beta(x^2 - 4)$ with $\alpha\beta = 12$.

2. Then $a = \alpha$, $b = -3\alpha$, $c = \beta$, $d = -4\beta$, and $a + b + c + d = -2\alpha - 3\beta$.

3. Test the positive pairs (α, β) with $\alpha\beta = 12$: $(1, 12) \rightarrow -38$; $(2, 6) \rightarrow -22$; $(3, 4) \rightarrow -18$; $(4, 3) \rightarrow -17$; $(6, 2) \rightarrow -18$; $(12, 1) \rightarrow -27$.

4. The least is -38, from $(x^2 - 3)(12x^2 - 48)$. Check: $(x^2 - 3)(12x^2 - 48) = 12x^4 - 48x^2 - 36x^2 + 144 = 12x^4 - 84x^2 + 144$ and $1 - 3 + 12 - 48 = -38$.

5. (Moving the whole 12 onto the first binomial gives $(12x^2 - 36)(x^2 - 4)$, sum -27, which is not the least.)

Answer: (D)

15.

SAT CHALLENGE · NO CALCULATOR NEEDED

The quadratic function f has its vertex at $(3, 4)$. The function g is defined by $g(x) = -2f(x+1) + 7$, and the graph of $y = g(x)$ passes through the point $(4, 2)$. What is the value of $f(9) - f(2)$?

1. Vertex form: $f(x) = a(x-3)^2 + 4$.

2. Use the point on g : $g(4) = -2f(5) + 7 = 2 \Rightarrow f(5) = \frac{5}{2}$. (The shift gives $f(4+1) = f(5)$.)

3. $f(5) = a(2)^2 + 4 = \frac{5}{2} \Rightarrow 4a = -\frac{3}{2} \Rightarrow a = -\frac{3}{8}$.

4. $f(9) = -\frac{3}{8}(36) + 4 = -\frac{27}{2} + 4 = -\frac{19}{2}$ and $f(2) = -\frac{3}{8}(1) + 4 = \frac{29}{8}$.

5. $f(9) - f(2) = -\frac{76}{8} - \frac{29}{8} = -\frac{105}{8}$. (Shortcut: the $+4$ cancels, so the difference is $a(36-1) = 35 \cdot \left(-\frac{3}{8}\right) = -\frac{105}{8}$.)

Answer: $-\frac{105}{8}$